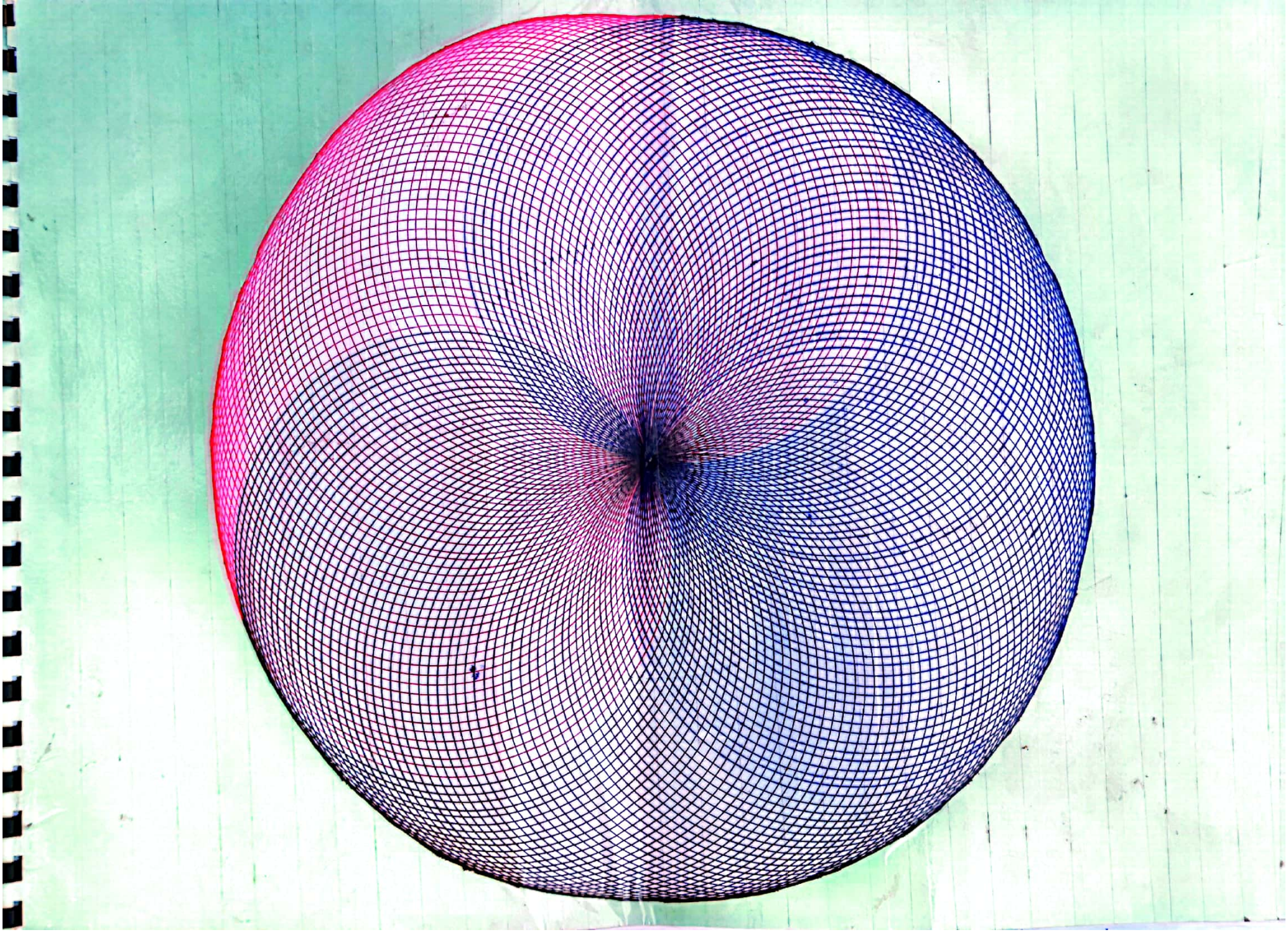
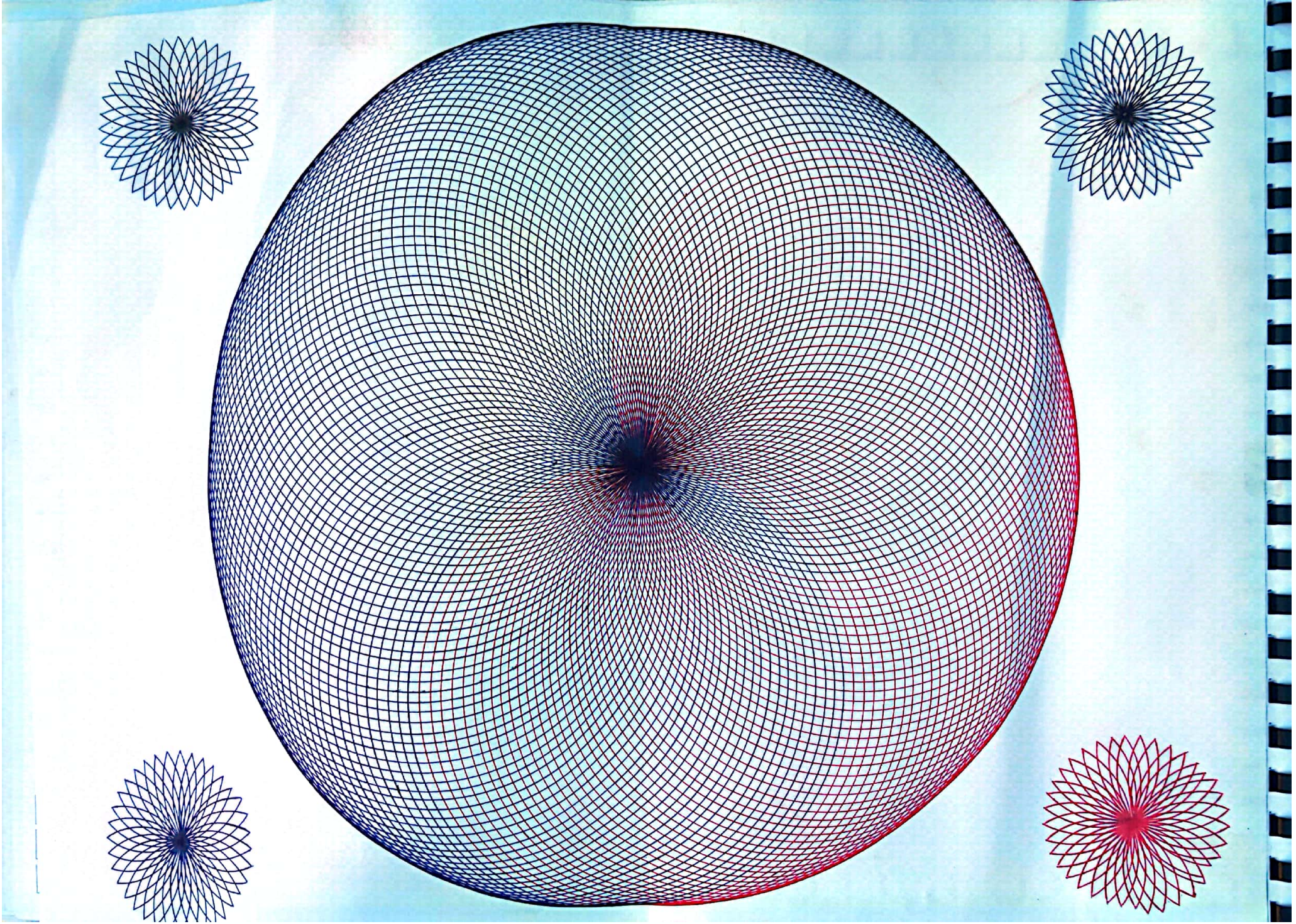


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COVER PAGE

NAME: ELIJAH MASEKA

EXAM NUMBER: 180400370009

CENTER NUMBER: 4011

CENTER NAME: CHIZONGWE STEM SECONDARY SCHOOL

PROJECT TITLE: TOOL BOX

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THEME

MANAGEMENT

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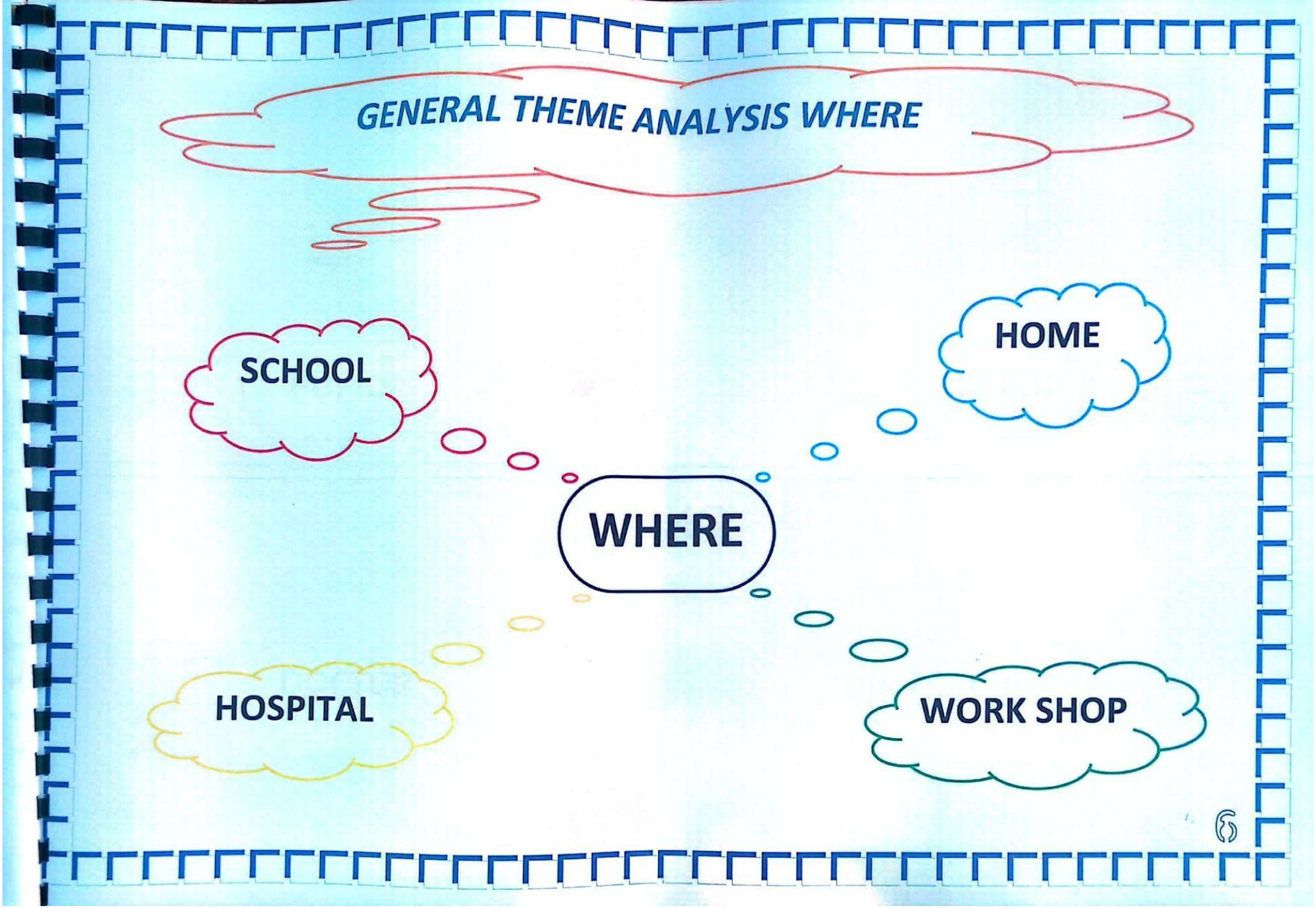
THEME DEFINITION

- Management is the best way to put things in place. (English Dictionary)
- Management is the art of getting things done through others and with formally organized groups. (Oxford Dictionary)
- Management is a unique process that consists of the actions of planning, organizing, and controlling performed to determine and achieve the goals that have been determined through the use of human resources and other resources. (Google)

WORKING DEFINITION

- Management is the handling of things in a place (Bank, Church or School) in an ordinary way.

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GENERAL THEME ANALYSIS WHO

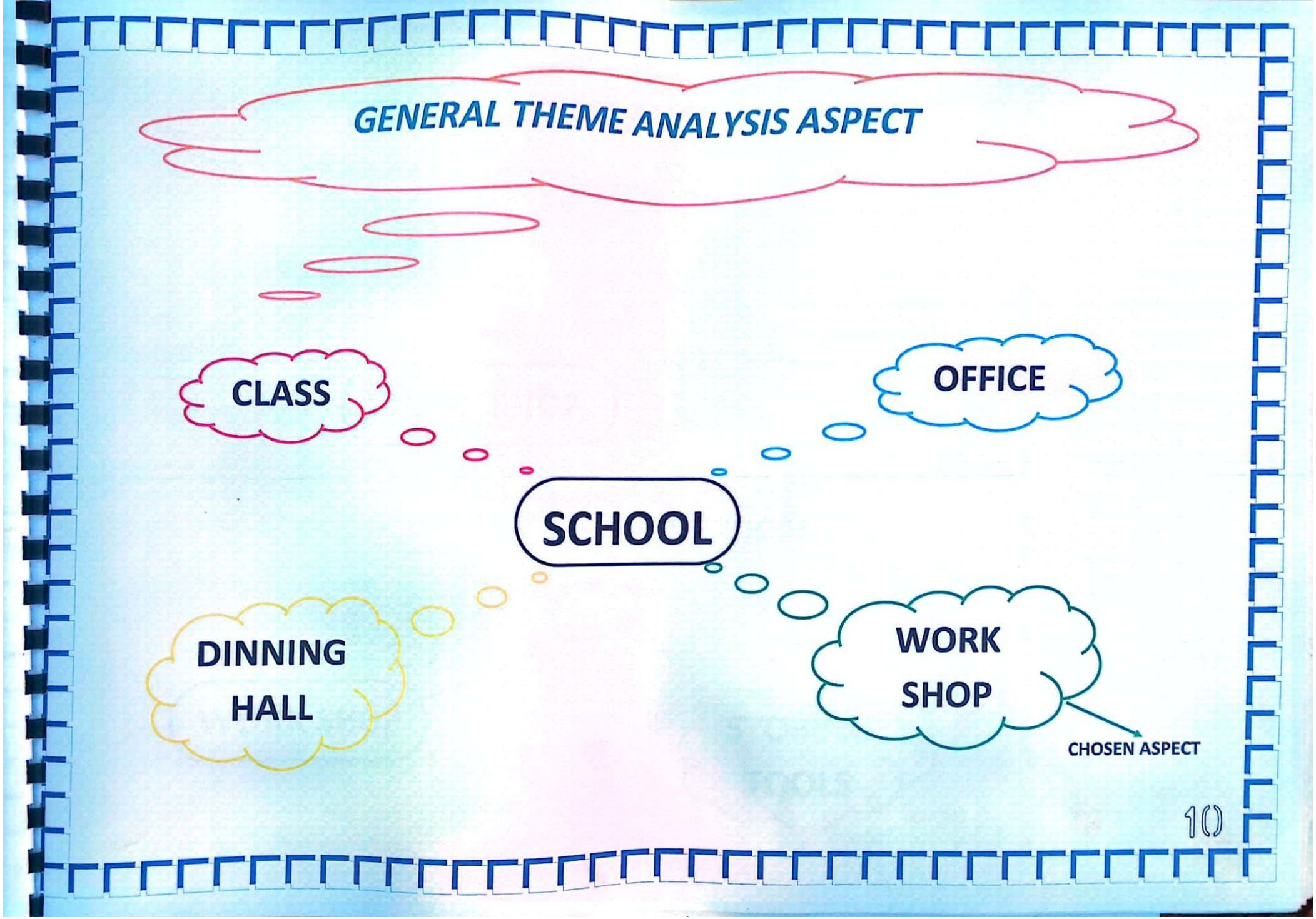


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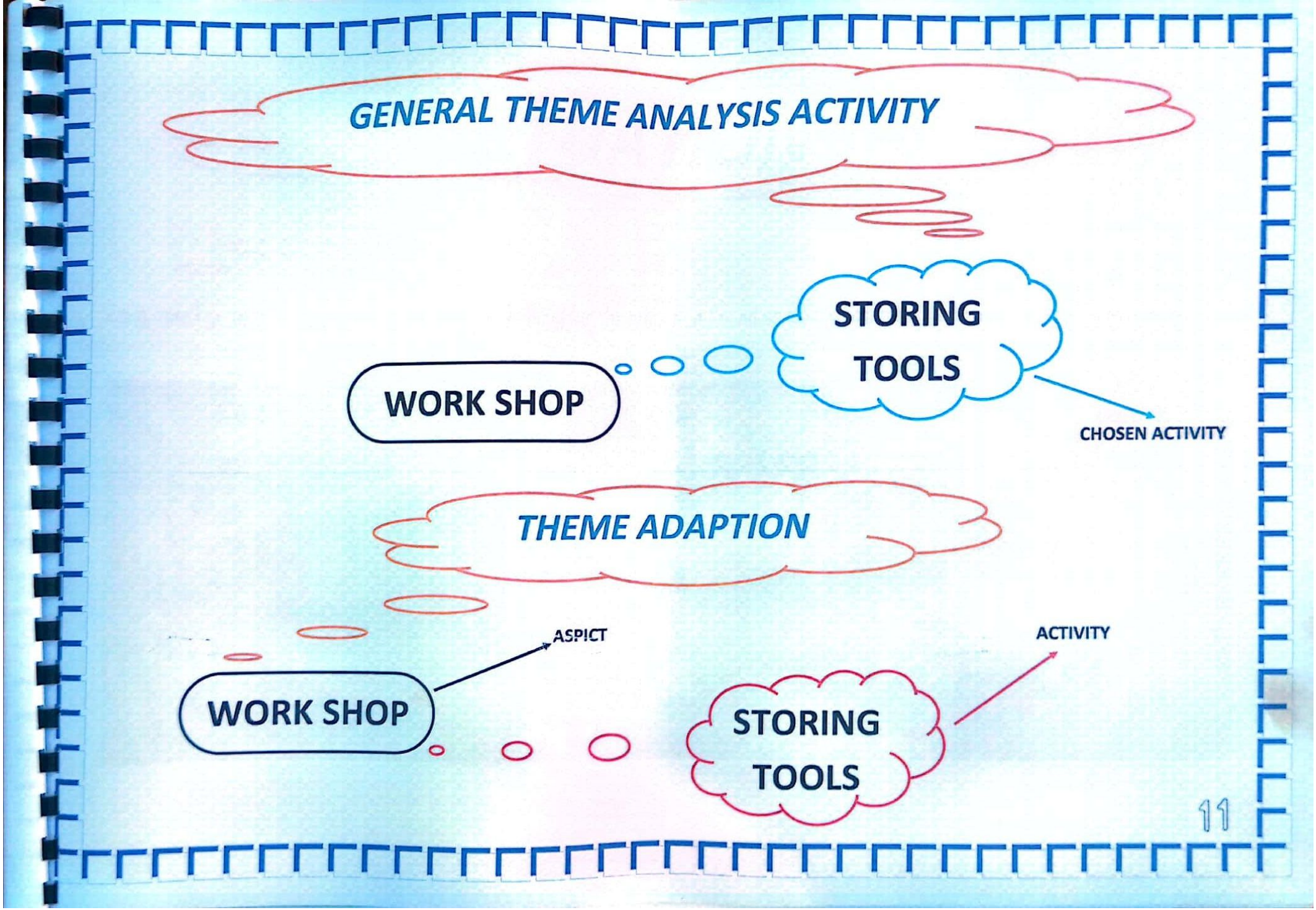
GENERAL THEME ANALYSIS WHY



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SITUATION

Each time I go to the work shop I find it difficulty looking for the tools that I need as they are scattered all over the work shop.



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PROBLEM IDENTIFICATION

There is no artifact or product which can be used to store tools in an orderly manner so that I can easily locate tools when I go to the work shop.

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DESIGN BRIEF

Design and make an artifact which can be used to store tools for easy locating in the work shop when I need them.

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RESEARCH

I carried out my research using the following research tools; Google, Observation, Interviews. And my finding were as following:

1) **Function:** The artifact is used for storing tools.
2) **Material used:** They usually use wood to make the project.
3) **Cost:** The artifact ranges between K500 to K800.
4) **User:** The artifact is used by people above the age of 14 years.
5) **Construction:** They usually use permanent joints.
6) **Time to make:** They take 4 work days of 3 hours each.
7) **Finish:** They usually use varnish to finish the artifact
8) **Safety:** They carve the edges.
9) **Size:** The artifact ranges between length 60cm, width 40cm, and height 45cm.

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EXCISTING IDEAS



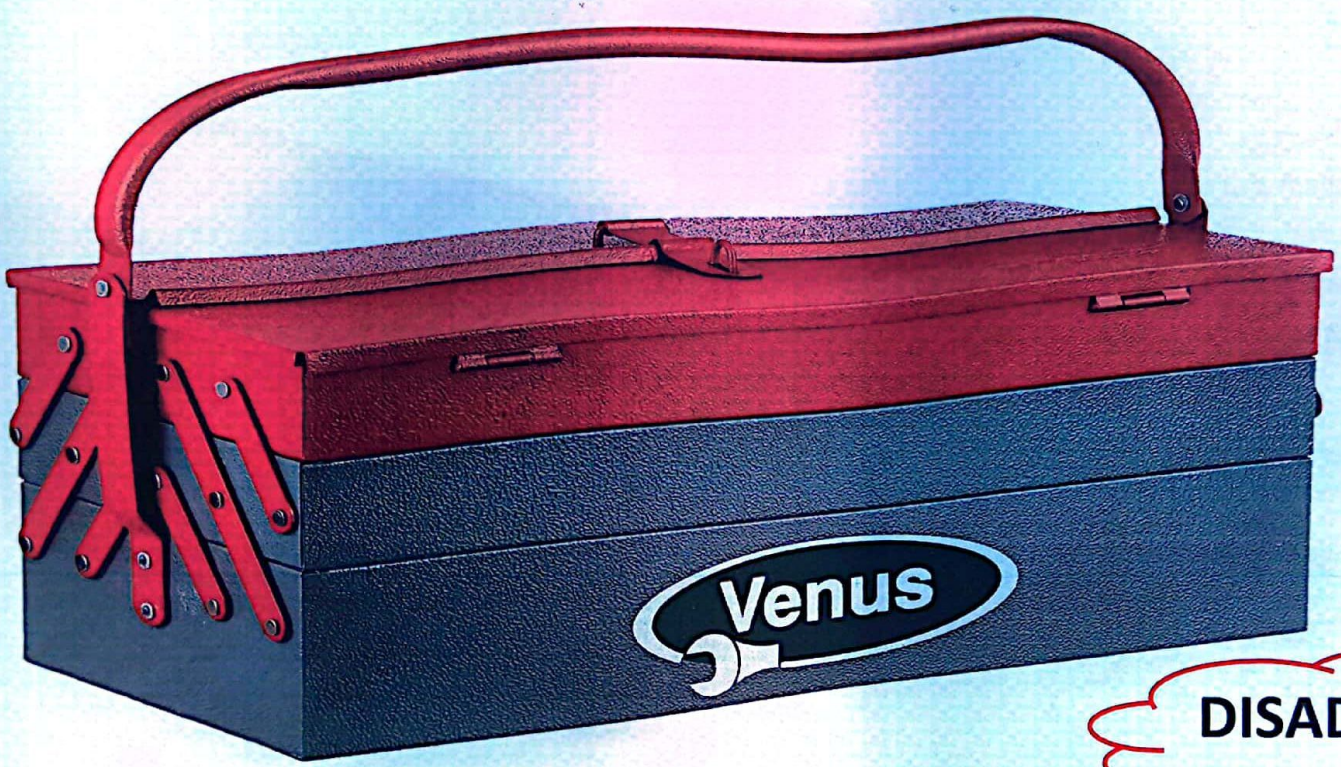
ADVANTAGES

It has wheels which makes it easy to move

SAFETY

It is made out of non-toxic material

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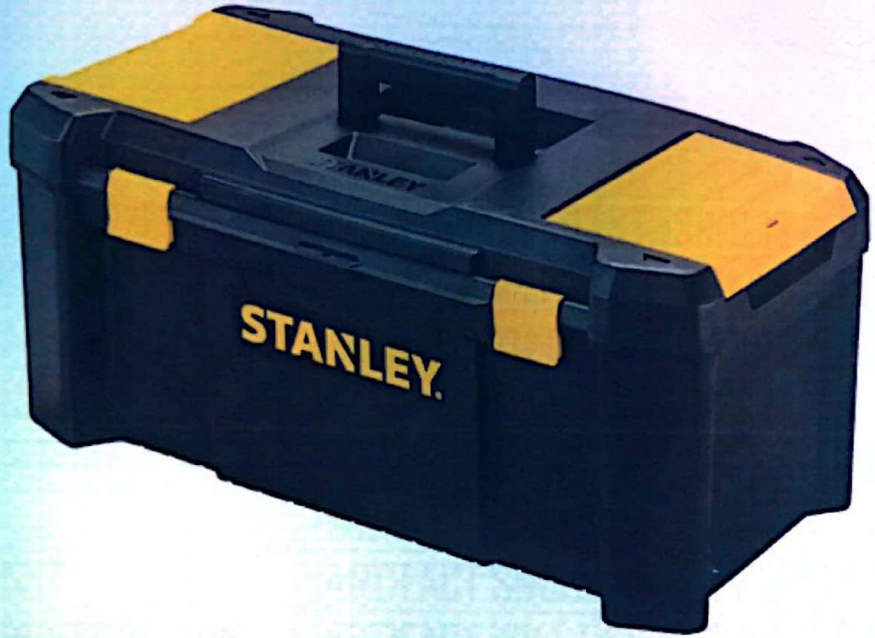
DISADVANTAGE

It is expensive

ADVANTAGE

It is strong

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SAFETY

It is made out of non- toxic material

MATERIAL

It is made out of plastic

DISADVANTAGE

It is not strong

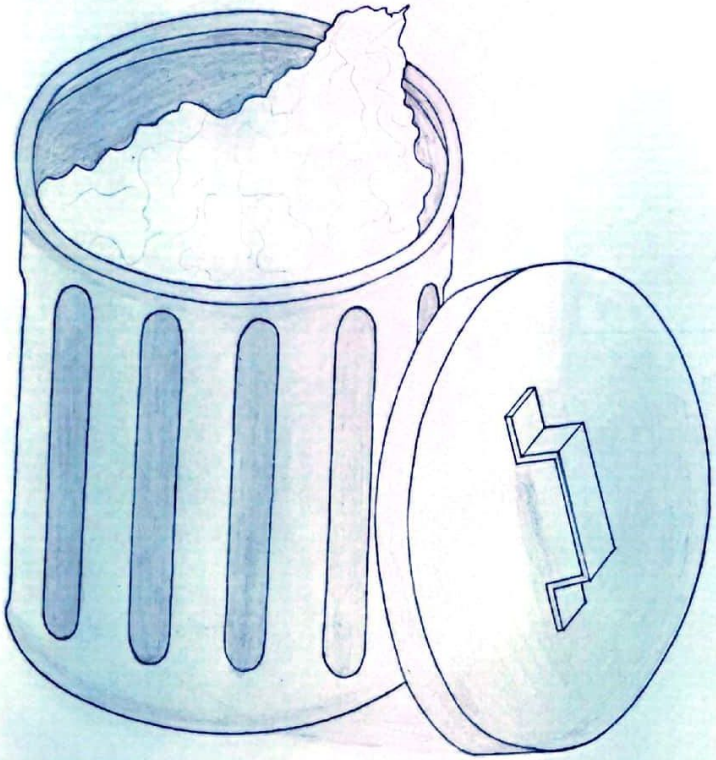
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SPECIFICATION

FUNCTIONS	TO BE USED FOR STORING TOOLS
MATERIALS	THE ARTIFACT MADE WILL USE MATERIAL WHICH CAN NOT EASILY GET DAMAGED
SAFETY	ARTIFACT TO BE MADE WITH NON-TOXIC MATERIALS
SIZE	HEIGHT: 45CM LENGTH: 60CM WIDTH: 40CM
COST	ARTIFACT SHOULD COST BETWEEN THE RANGE OF K500TO K800
CONSTRUCTION	ARTIFACT WILL USE PERMANENT JOINTS WHICH ARE STRONG AND CAN LAST FOR A LONG TIME
APPEARANCE	ARTIFACT SHOULD LOOK ATTRACTIVE
USER	ARTIFACT WILL BE USED BY PEOPLE ABOVE THE AGE OF 14 YEARS
FINISH	WILL USE VERNISH WHICH WILL MAKE IT LOOK ATRACTIVE

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POSSIBLE SOLUTION 1



ADVANTAGES

It is easy to work with

It looks attractive

It is strong

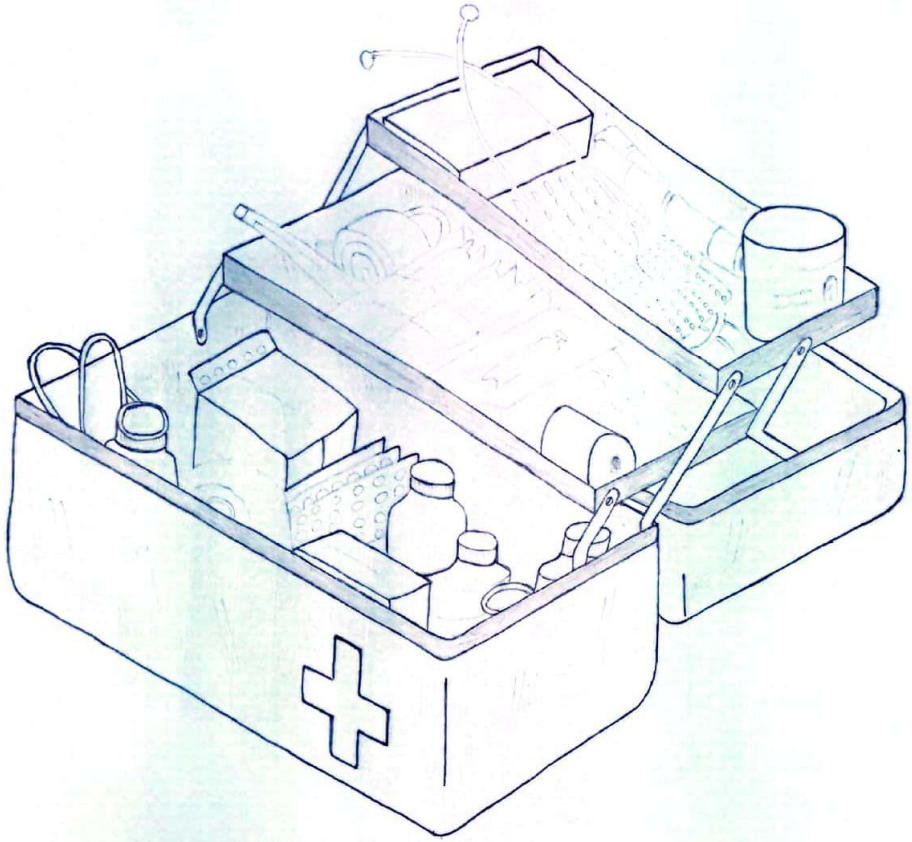
DISADVANTAGES

It requires skilled labor

It is expensive

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POSSIBLE SOLUTION 2



ADVANTAGES

It is easy to work with

It is light in weight

It looks attractive

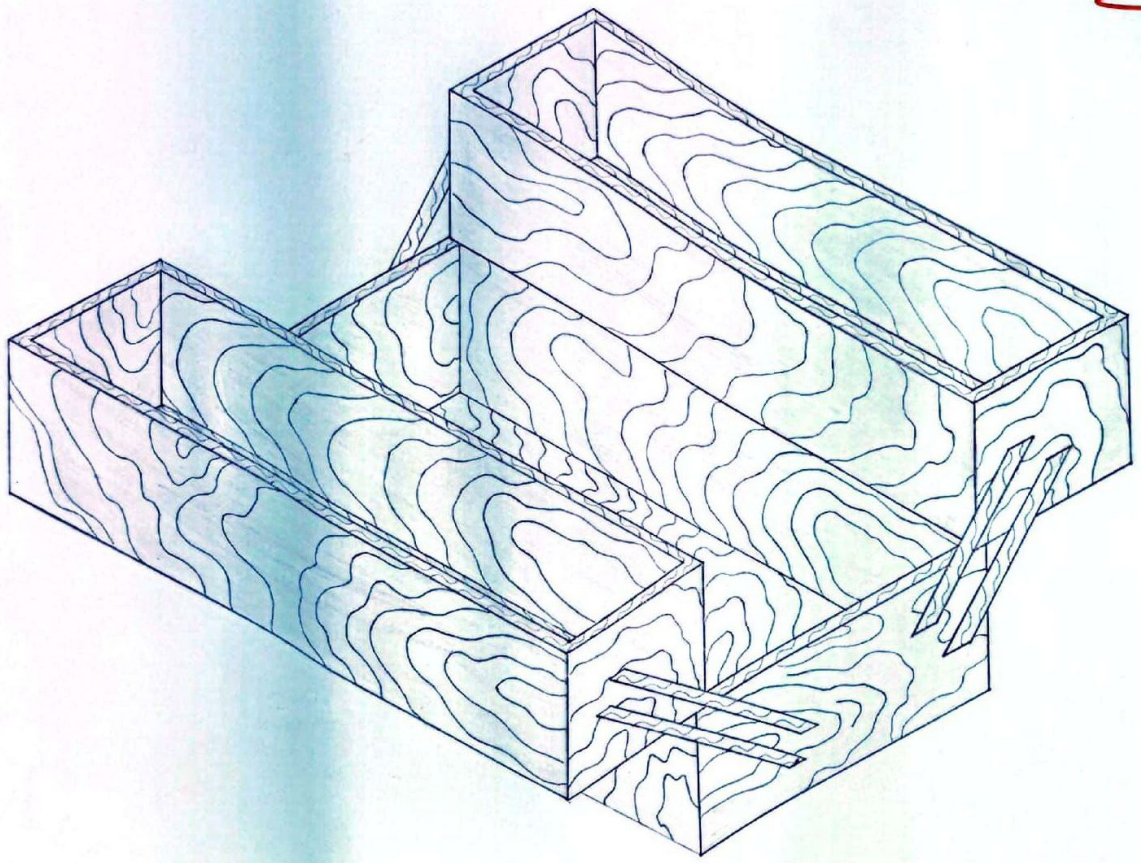
DISADVAUTAGES

It requires skilled labor

It is expensive

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POSSIBLE SOLUTION 3



ADVANTAGE

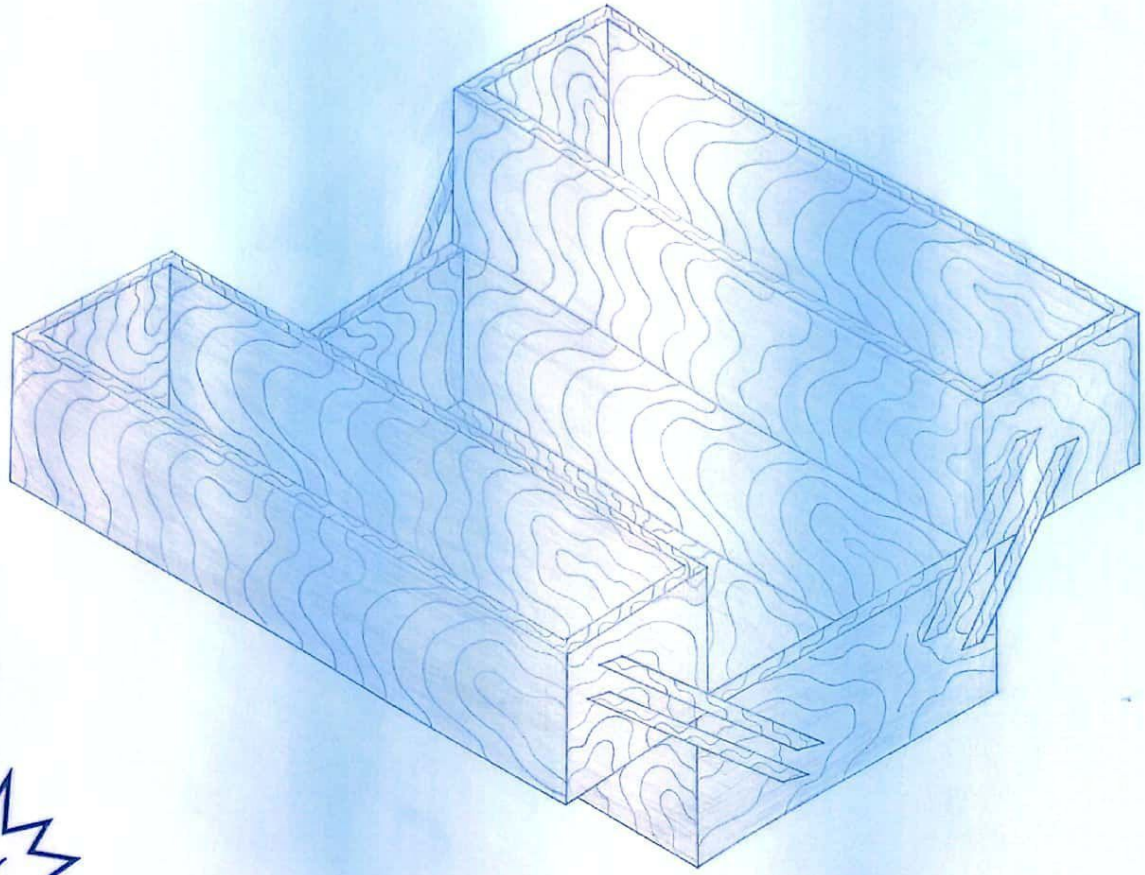
It is easy to work with

DISADVANTAGE

It requires skilled labor

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CHOSEN SOLUTION



FUNCTION

To store tools in the work shop

ADVANTAGE

It is easy to work with

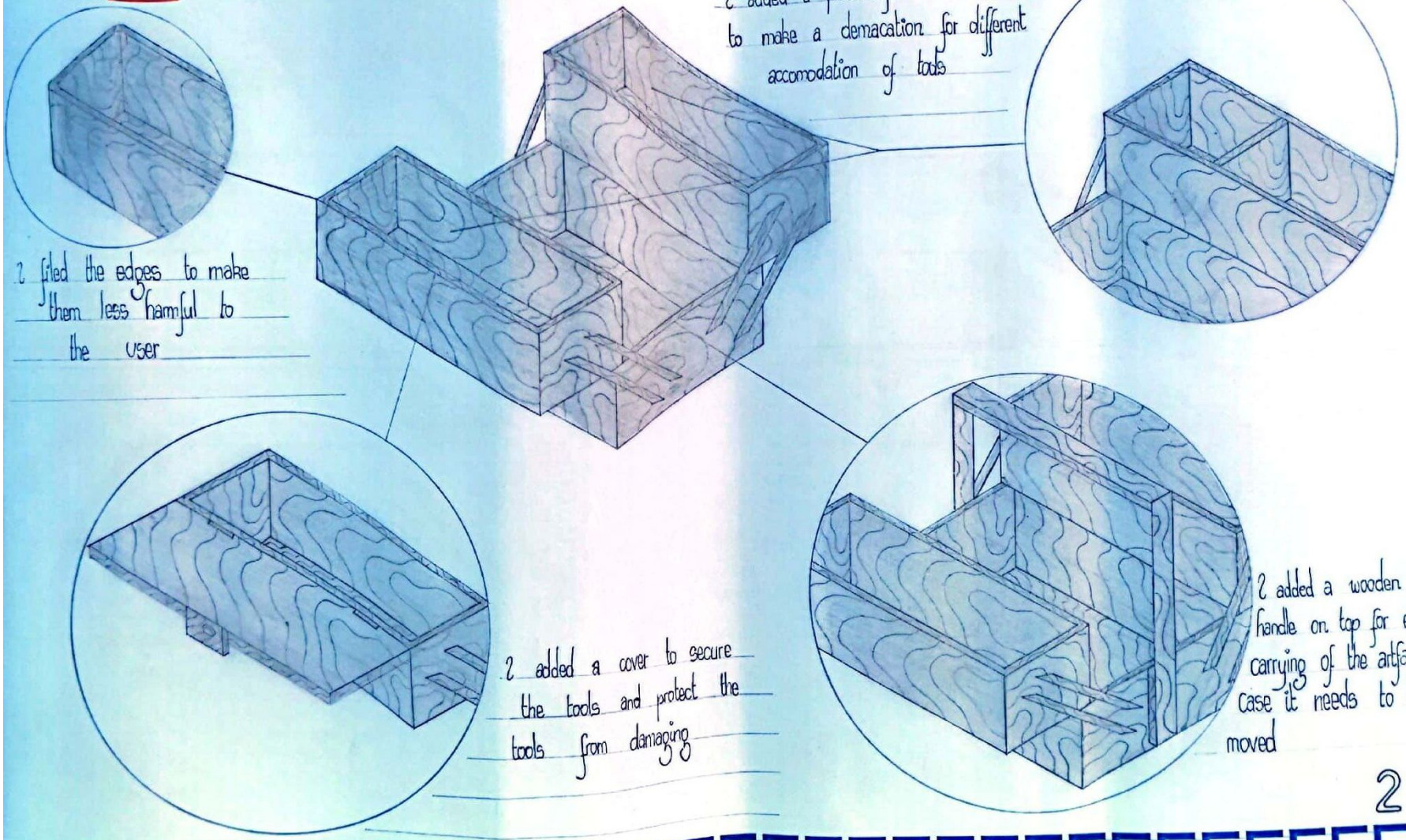
DISADVANTAGES

It requires skilled labor

It is expensive

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DEVELOPMENT OF CHOSEN IDEA



I filed the edges to make them less harmful to the user

I added a piece of wood in between to make a demacation for different accomodation of tools

I added a cover to secure the tools and protect the tools from damaging

I added a wooden handle on top for easy carrying of the artfact in case it needs to be moved

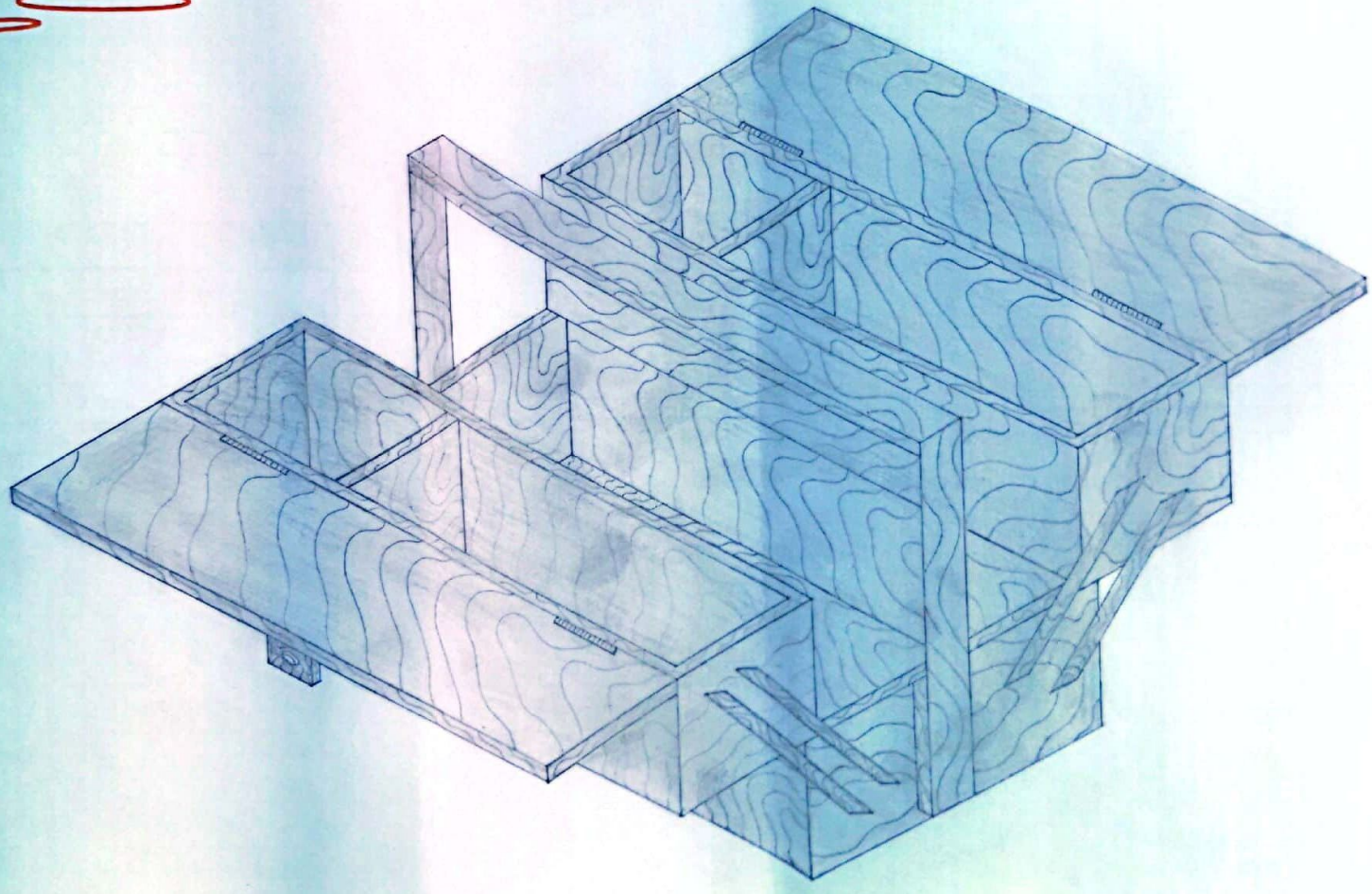
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MATERIAL LIST

S/N	DESCRIPTION	QUANTITY
1	Hard wood	8
2	Varnish	1 Liter
3	Nails	30
4	Screws	18

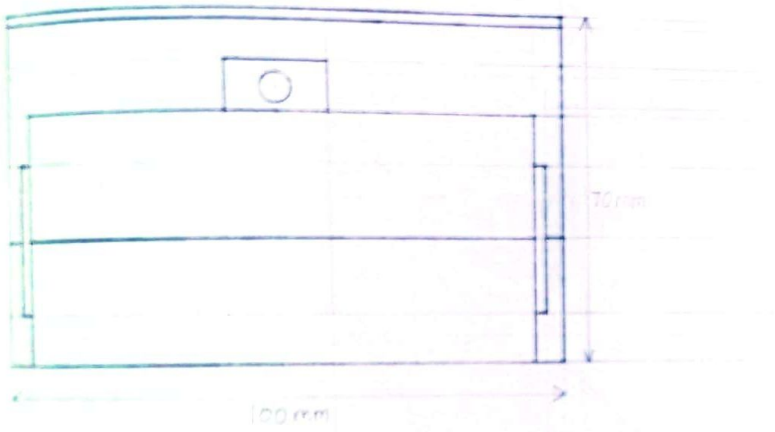
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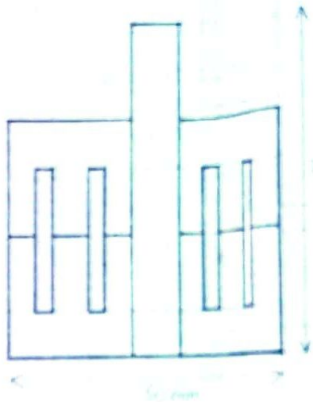
PRESENTATION DRAWING

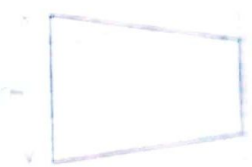


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WORKING DRAWING











NAME : MASEKA ELIJAH
EXAM NUMBER : 180400376007
CENTER NUMBER : 4011
CENTER NAME : CHIZONGWE
PROJECT TITLE : TOOL BOX
SCALE : 1:8
CHECKED BY :

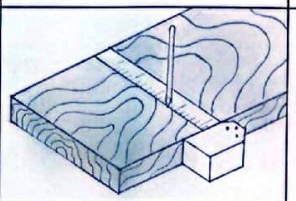
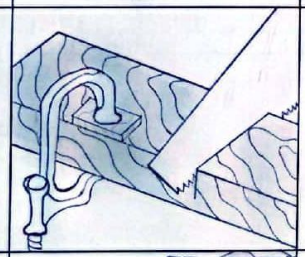
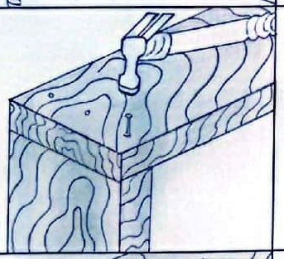
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CUTTING LIST

PART	QUANTITY	MATERIAL	LENGTH	WIDTH
Hundle	1	Hard wood	60cm	40cm
Front and rare end	2	Hard wood	34cm	20cm
Side Board	2	Hard wood	40cm	30cm
Base	1	Hard wood	55cm	40cm

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PRODUCTION PLAN

STAGE	PROCESS	SKETCHES	TOOLS	ESTIMATED TIME	ACTUAL TIME	COMMENTS
1.	Measuring and marking out		Try square and pencil	10 minutes	20 minutes	It took more time than I estimated
2.	Cutting		Hacksaw	20 minutes	30 minutes	Cutting took me time than I estimated because I didn't have much experience
3.	Nailing		Hammer	15 minutes	25 minutes	Nailing took time because I was trying to position the nails at ninety degrees angles at first
4.	Finishing		Varnishing	30 minutes	40 minutes	I took much time because I was slow

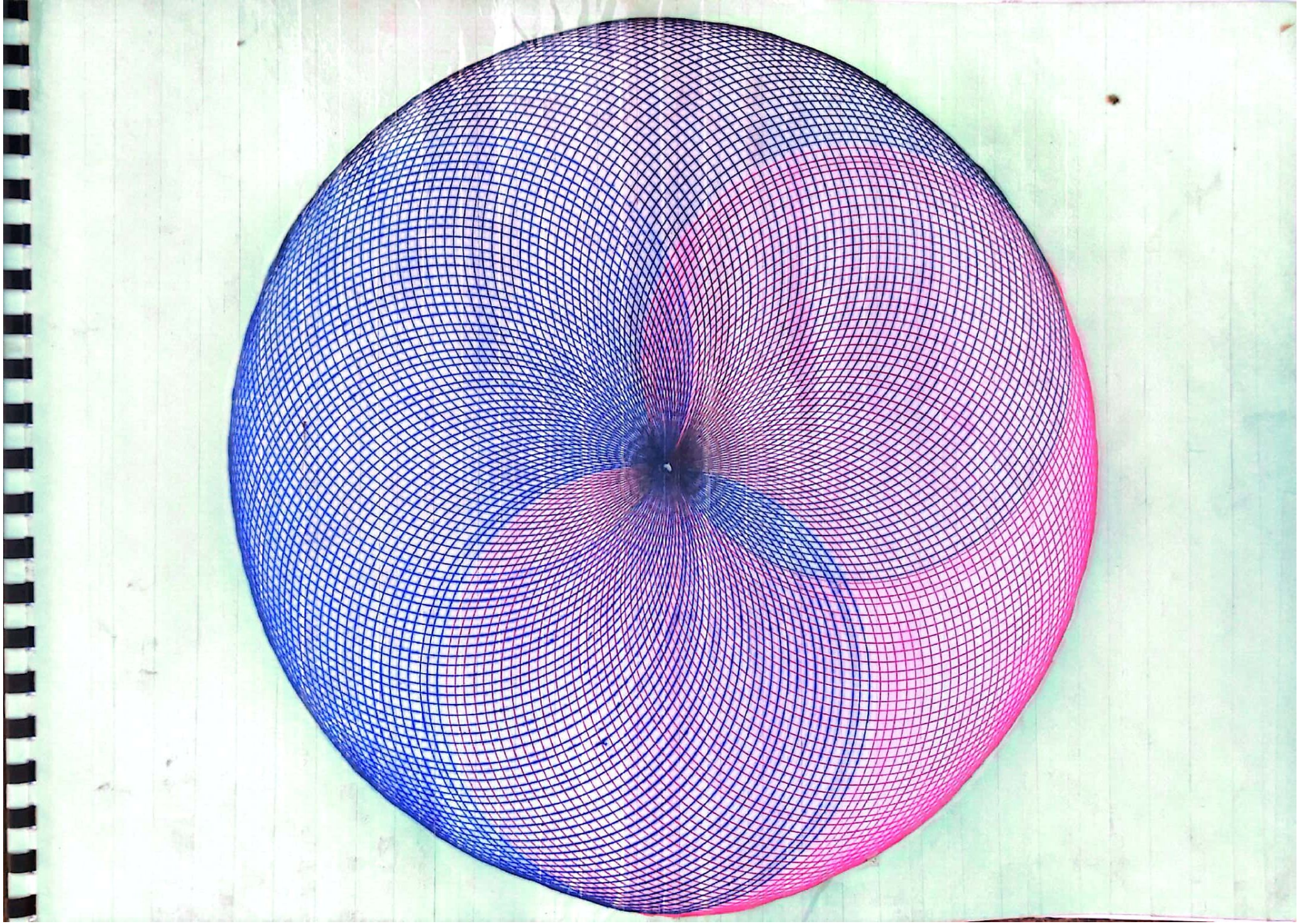
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EVALUATION

STRENGTH	WEAKNESS	COST
• The artifact has a stable condition • The materials used were strong and durable • The appearance looks attractive	• The material used may rust • It is heavy • It takes a lot of time to make	• The artifact is very expensive

CHANGES MADE	FUTURE IMPROVEMENTS
• Changed the design • Changed the material and the finish	• I would use soft wood to make it light in weight • I would use simple construction which doesn't take a lot of time • I would look for cheap but attractive material to make my artfact less expensive

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